Marwan Haithem Amin

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Education

Qatar University **Aug. 2021 – Jul. 2026**
Bachelor of Science in Computer Engineering *GPA: 4.00 / 4.00*

- Minor: Engineering Project Management
- Academic Awards: Vice President's Lists and Dean's Lists (2021 – 2026)
- Selected Electives: Reinforcement Learning, Computer Vision, Wireless Networks & Applications, Computer Security

Research & Professional Experience

Qatar Computing Research Institute (QCRI) **May 2026 – Sept. 2026**
Research Intern – Federated Learning and Federated Unlearning for Healthcare AI *Doha, Qatar*

- Developed a privacy-preserving oral cancer classification framework using Python, PyTorch, Flower, and ResNet-18, evaluating federated learning across three hospital datasets comprising 3,175 medical images.
- Implemented and benchmarked 4 Federated Learning algorithms (FedAvg, FedProx, FedBN, and FedMOON) under non-IID data distributions, achieving 76.45% test accuracy with the FedAvg baseline.
- Implemented client-level Federated Unlearning techniques, including FUSED-based sparse adapters and Low-Rank Adaptation (LoRA), alongside full retraining baselines to evaluate selective knowledge removal.
- Applied parameter-efficient unlearning techniques, reducing trainable parameters to approximately 2.22% using LoRA, and investigated critical-layer identification for selective model updates.
- Developed GPU-based experimentation and evaluation pipelines using SLURM, Ray, and TensorBoard to analyze model accuracy, knowledge retention, forgetting effectiveness, and relearning behavior across hospitals.

Qatar University **Jan. 2025 – Jun. 2026**
Tutor and Administrator – Engineering Success Oasis *Doha, Qatar*

- Provided academic tutoring to engineering students across multiple core subjects, including Data Structures, Operating Systems, Signals and Systems, and Electric Circuits, through both one-to-one sessions and laboratory instruction.
- Explained complex engineering concepts in a clear and structured manner to improve student understanding and performance.
- Assisted students with assignments, problem solving, and exam preparation through personalized academic support.
- Certified by the National Tutoring Association (NTA), ensuring adherence to professional tutoring standards.

FIFA 2022 World Cup **Oct. 2022 – Dec. 2022**
SPS Supervisor *Doha, Qatar*

- Provided on-site assistance and guidance to international spectators during a large-scale global sporting event.
- Managed crowd flow, supported access control, and delivered accurate event information in a fast-paced, multicultural environment.
- Developed strong communication, teamwork, and problem-solving skills.

Technical Skills

Programming: Python, Java, Bash, MATLAB
AI & ML: PyTorch, Computer Vision, Deep Learning, Reinforcement Learning, Federated Learning, Federated Unlearning
Networking: TCP/IP, Routing & Switching, VLANs, Network Troubleshooting, Network Monitoring
Tools & Frameworks: OpenCV, Flower, Docker, Prometheus, Grafana, Git, GitHub, YOLOv11
Embedded Systems: Raspberry Pi, Arduino, ESP32, Sensors & Actuators
Systems: Linux, Windows, Data Structures, Operating Systems

Projects

Senior Design Project – RoboDog (HARIS) **Aug. 2025 – Jun. 2026**

- Developed HARIS (Hazard Autonomous Robotic Intervention System), an autonomous quadruped robot designed to patrol oil and gas facilities, detect gas leaks, locate shut-off valves, and autonomously close them to reduce human exposure to hazardous environments.
- Integrated MQ-2/MQ-4 gas sensors, DHT11 environmental sensing, onboard computer vision, and wireless monitoring for real-time leak detection.
- Trained a YOLOv11m model for valve-state detection, achieving AUC scores of 0.995 for closed valves and 0.986 for open valves.
- Applied Soft Actor-Critic (SAC) deep reinforcement learning to control a custom 6-DOF robotic arm, achieving 100% reaching and 97% valve-rotation success in simulation.

Remote Network Monitoring System (RNMS-PR) **May 2024**

- Developed a Raspberry Pi-based network monitoring solution using Prometheus and Grafana to collect, analyze, and visualize real-time network performance metrics.
- Configured Kali Linux and a Prometheus server, integrating Node, SNMP, and Blackbox Exporters for monitoring network traffic, connectivity, packet loss, and system health.
- Integrated ZeroTier for secure remote access and an Apache-hosted web interface for monitoring dashboards.
- Coordinated project planning, implementation milestones, technical documentation, and system evaluation across different network environments.

Reacher Arm Control **May 2025**

- Implemented reinforcement learning algorithms to train agents for control and decision-making tasks.
- Designed reward functions and evaluated agent performance using simulation-based analysis.

Object Detection and Tracking for Football Matches **Nov. 2024**

- Developed image processing pipelines for object detection and visual analysis.
- Applied feature extraction and classification techniques using Python.

Professional Certifications

Cisco Certified Network Associate (CCNA) **Oct. 2026**
Cisco Systems — Expires: Oct. 2029

SAP Certified – Implementation Consultant – SAP S/4HANA Cloud Private Edition **Aug. 2026**
SAP — Expires: Aug. 2027

Courses & Professional Training

SAP Dual Study Program **Jun. 2026 – Jul. 2026**
SAP

Cybersecurity Fundamentals **Jun. 2024**
IBM SkillsBuild

Artificial Intelligence Fundamentals **Jun. 2024**
IBM SkillsBuild